

BLINKIT GROCERY SALES ANALYSIS

SQL Data Cleaning & Exploratory Data Analysis

Project Detail	Information
Tool	MySQL / MySQL Workbench
Dataset	Blinkit Grocery Sales Dataset
Records Analyzed	8,523
Objective	Data cleaning, validation and SQL-based business analysis

Beginner-level SQL Portfolio Project

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1. Project Overview

This project analyzes a Blinkit grocery sales dataset using MySQL. The project focuses on data import, data quality checks, data cleaning, validation and SQL-based exploratory analysis.

Workflow: Raw Data → Import → Data Quality Checks → Data Cleaning → Validation → SQL Analysis → Business Insights

2. Dataset Description

The dataset contains product-level and outlet-level attributes along with item outlet sales. The main fields include product identifier, weight, fat content, visibility, product type, MRP, outlet identifier, establishment year, outlet size, location type, outlet type and item outlet sales.

Column	Description
Item_Identifier	Product identifier
Item_Weight	Product weight
Item_Fat_Content	Product fat-content category
Item_Visibility	Product visibility measure
Item_Type	Product category/type
Item_MRP	Maximum Retail Price
Outlet_Identifier	Outlet identifier
Outlet_Establishment_Year	Outlet establishment year
Outlet_Size	Outlet size
Outlet_Location_Type	Outlet location tier
Outlet_Type	Outlet type
Item_Outlet_Sales	Item sales value at the outlet

A	B	C	D	E	F	G	H	I	J	K	L
Item_Identifier	Item_Weight	Item_Fat_Content	Item_Visibility	Item_Type	Item_MRP	Outlet_Identifier	Outlet_Establishment_Year	Outlet_Size	Outlet_Location_Type	Outlet_Type	Item_Outlet_Sales
FDA15	9.3	Low Fat	0.016047301	Dairy	249.8092	OUT049	1999	Medium	Tier 1	Supermarket Type1	3735.
DR001	5.92	Regular	0.019278216	Soft Drinks	48.2692	OUT018	2009	Medium	Tier 3	Supermarket Type2	443.4
FDN15	17.5	Low Fat	0.016760075	Meat	141.618	OUT049	1999	Medium	Tier 1	Supermarket Type1	2097
FDX07	19.2	Regular		0 Fruits and Vegetables	182.095	OUT010	1998		Tier 3	Grocery Store	732
NCD19	8.93	Low Fat		0 Household	53.8614	OUT013	1987	High	Tier 3	Supermarket Type1	994.7
FDP36	10.395	Regular		0 Baking Goods	51.4008	OUT018	2009	Medium	Tier 3	Supermarket Type2	556.6
FDO10	13.65	Regular	0.012741089	Snack Foods	57.6588	OUT013	1987	High	Tier 3	Supermarket Type1	343.5
FDP10		Low Fat	0.127469857	Snack Foods	107.7622	OUT027	1985	Medium	Tier 3	Supermarket Type3	4022.7
FDH17	16.2	Regular	0.016687114	Frozen Foods	96.9726	OUT045	2002		Tier 2	Supermarket Type1	1076.5
FDU28	19.2	Regular	0.09444959	Frozen Foods	187.8214	OUT017	2007		Tier 2	Supermarket Type1	4710.
FDV07	11.8	Low Fat		0 Fruits and Vegetables	45.5402	OUT049	1999	Medium	Tier 1	Supermarket Type1	1516.0
FDA03	18.5	Regular	0.045463773	Dairy	144.1102	OUT046	1997	Small	Tier 1	Supermarket Type1	2187.
FDX32	15.1	Regular	0.1000135	Fruits and Vegetables	145.4786	OUT049	1999	Medium	Tier 1	Supermarket Type1	1589.2
FDX46	17.6	Regular	0.047257328	Snack Foods	119.6782	OUT046	1997	Small	Tier 1	Supermarket Type1	2145.2
FDI32	16.35	Low Fat	0.0680243	Fruits and Vegetables	196.4426	OUT013	1987	High	Tier 3	Supermarket Type1	1977.
FDI49	9	Regular	0.069088961	Breakfast	56.3614	OUT046	1997	Small	Tier 1	Supermarket Type1	1547.3
NCB42	11.8	Low Fat	0.008596051	Health and Hygiene	115.3492	OUT018	2009	Medium	Tier 3	Supermarket Type2	1621.8
FDI49	9	Regular	0.069196376	Breakfast	54.3614	OUT049	1999	Medium	Tier 1	Supermarket Type1	718.3
DR111		Low Fat	0.034237682	Hard Drinks	113.2834	OUT027	1985	Medium	Tier 3	Supermarket Type3	2303.
FDU02	13.35	Low Fat	0.10249212	Dairy	230.5352	OUT035	2004	Small	Tier 2	Supermarket Type1	2748.4
NDN22	18.85	Regular	0.138190277	Snack Foods	250.8724	OUT013	1987	High	Tier 3	Supermarket Type1	3775.
FDW12		Regular	0.035399923	Baking Goods	144.5444	OUT027	1985	Medium	Tier 3	Supermarket Type3	4064.0
NCB30	14.6	Low Fat	0.025698134	Household	196.5084	OUT035	2004	Small	Tier 2	Supermarket Type1	1587.2
FDI37		Low Fat	0.057556998	Baking Goods	107.6938	OUT019	1985	Small	Tier 1	Grocery Store	214.3
FDI28	13.85	Regular	0.025896485	Frozen Foods	165.021	OUT046	1997	Small	Tier 1	Supermarket Type1	4078.
NCDO6	13	Low Fat	0.099887103	Household	45.906	OUT017	2007		Tier 2	Supermarket Type1	838.
FDV10	7.645	Regular	0.066693437	Snack Foods	42.3112	OUT035	2004	Small	Tier 2	Supermarket Type1	1065
DR159	11.65	low fat	0.019356132	Hard Drinks	39.1164	OUT013	1987	High	Tier 3	Supermarket Type1	308.9
FDI51	5.925	Regular	0.161466534	Dairy	45.5086	OUT010	1998		Tier 3	Grocery Store	178.4
FDI14		Regular	0.072221801	Canned	43.6454	OUT019	1985	Small	Tier 1	Grocery Store	125.8
FDV38	19.25	Low Fat	0.170348551	Dairy	55.7956	OUT010	1998		Tier 3	Grocery Store	163.7
NCI57	18.6	Low Fat	0.080829372	Health and Hygiene	96.4436	OUT018	2009	Medium	Tier 3	Supermarket Type2	2741.7
FDP33	18.7	Low Fat		0 Snack Foods	256.6672	OUT018	2009	Medium	Tier 3	Supermarket Type2	3068.0
FDI23	17.85	Low Fat		0 Breads	93.1436	OUT045	2002		Tier 2	Supermarket Type1	2174.5
DRH01	17.5	Low Fat	0.097904029	Soft Drinks	174.8738	OUT046	1997	Small	Tier 1	Supermarket Type1	2085.2
NCX29	10	Low Fat	0.089291137	Health and Hygiene	146.7102	OUT049	1999	Medium	Tier 1	Supermarket Type1	3791.0
FDV20		Regular	0.059511812	Fruits and Vegetables	128.0678	OUT027	1985	Medium	Tier 3	Supermarket Type3	2797.6

Original dataset opened in Excel

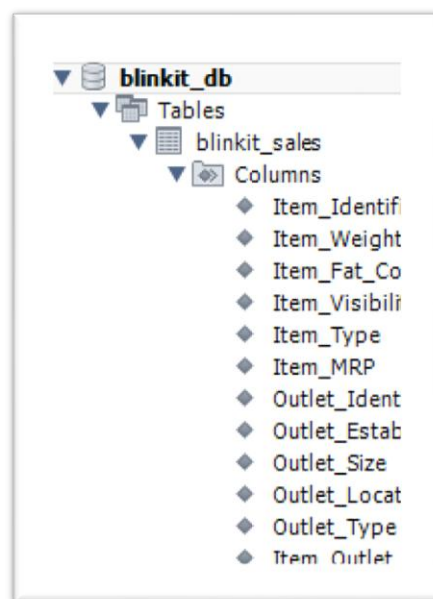
3. Tools and Technologies

Tool	Purpose
Excel / CSV	Source data and inspection
MySQL	Database and SQL analysis
MySQL Workbench	Import, queries, validation and results

4. Database Setup and Data Import

A dedicated database named blinkit_db was created for the project and the source CSV was imported into MySQL Workbench.

```
CREATE DATABASE IF NOT EXISTS blinkit_db;  
USE blinkit_db;
```



blinkit_db schema visible in MySQL Workbench

The CSV import screen reported 7,060 records imported into the initial imported table. The final analysis table blinkit_sales contained 8,523 records during the subsequent workflow.



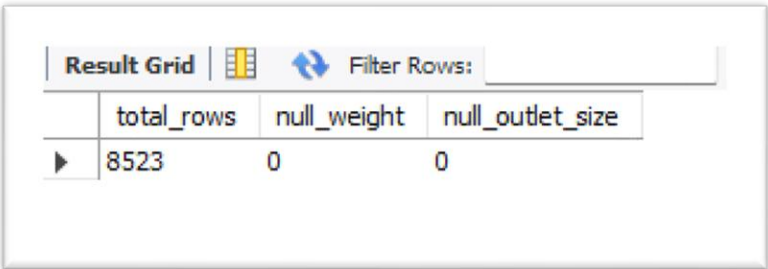
CSV import completion screen showing 7,060 records

5. Data Quality Checks

Record counts and missing-value checks were performed before analysis. The final table contained 8,523 records. Initial NULL checks for Item_Weight and Outlet_Size returned zero NULL values.

```
SELECT COUNT(*) AS total_rows
FROM blinkit_sales_raw;

SELECT COUNT(*) AS total_rows,
       SUM(Item_Weight IS NULL) AS missing_item_weight,
       SUM(Outlet_Size IS NULL) AS missing_outlet_size
FROM blinkit_sales_raw;
```

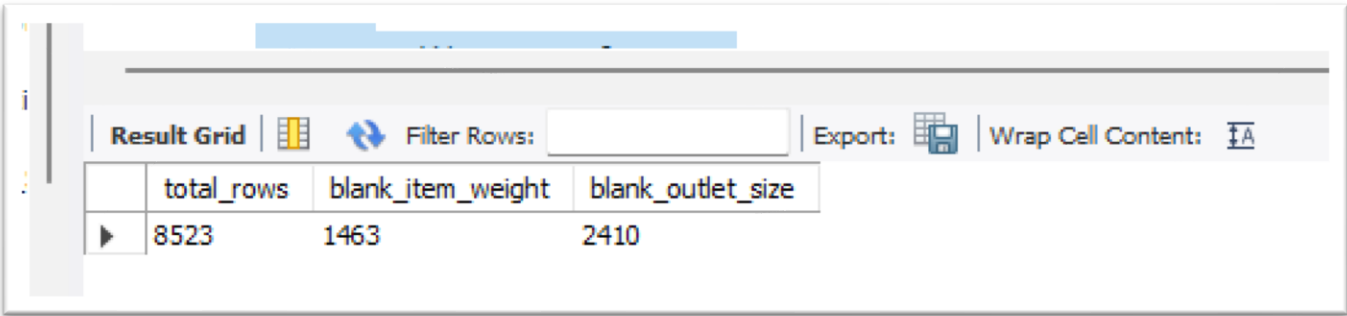


	total_rows	null_weight	null_outlet_size
▶	8523	0	0

Row-count and NULL-check result

A separate blank-value check identified 1,463 blank Item_Weight records and 2,410 blank Outlet_Size records in the raw data.

```
SELECT COUNT(*) AS total_rows,
       SUM(Item_Weight = "") AS blank_item_weight,
       SUM(Outlet_Size = "") AS blank_outlet_size
FROM blinkit_sales_raw;
```



	total_rows	blank_item_weight	blank_outlet_size
▶	8523	1463	2410

Blank-value check showing 1,463 and 2,410

6. Data Cleaning and Standardization

An inconsistency was identified in Item_Fat_Content. The same categories appeared under Low Fat, LF, Regular and reg.

Original Value	Standardized Value
----------------	--------------------

Low Fat	Low Fat
LF	Low Fat
Regular	Regular
reg	Regular

```
SET SQL_SAFE_UPDATES = 0;
```

```
UPDATE blinkit_sales
```

```
SET Item_Fat_Content =
```

```
  CASE
```

```
    WHEN Item_Fat_Content = 'LF' THEN 'Low Fat'
```

```
    WHEN Item_Fat_Content = 'reg' THEN 'Regular'
```

```
    ELSE Item_Fat_Content
```

```
  END
```

```
WHERE Item_Fat_Content IN ('LF', 'reg');
```

```
SET SQL_SAFE_UPDATES = 1;
```

Result Grid				
Filter Rows:				
Export: Export				
Wrap Cell Content: Wrap				
	Item_Fat_Content	total_records	total_sales	avg_sales
▶	Low Fat	5517	11904094.53	2157.71
	Regular	3006	6687030.88	2224.56

Final standardized Low Fat / Regular result

7. Data Type Validation

The final table structure was validated using DESCRIBE. Numeric fields such as Item_Weight, Item_Visibility, Item_MRP, Outlet_Establishment_Year and Item_Outlet_Sales use appropriate numeric types, while categorical fields use VARCHAR.

```
DESCRIBE blinkit_sales;
```

Result Grid Filter Rows: Export: Wrap Cell Content:						
	Field	Type	Null	Key	Default	Extra
▶	Item_Identifier	varchar(10)	YES		NULL	
	Item_Weight	decimal(10,3)	YES		NULL	
	Item_Fat_Content	varchar(20)	YES		NULL	
	Item_Visibility	decimal(10,8)	YES		NULL	
	Item_Type	varchar(50)	YES		NULL	
	Item_MRP	decimal(10,3)	YES		NULL	
	Outlet_Identifier	varchar(10)	YES		NULL	
	Outlet_Establishment_Year	int	YES		NULL	
	Outlet_Size	varchar(20)	YES		NULL	
	Outlet_Location_Type	varchar(20)	YES		NULL	
	Outlet_Type	varchar(30)	YES		NULL	
	Item_Outlet_Sales	decimal(12,4)	YES		NULL	

DESCRIBE blinkit_sales result

8. Exploratory Data Analysis

8.1 Overall KPIs

KPI	Value
Total Records	8,523
Total Sales	18,591,125.41
Average Sales per Record	2,181.29

Result Grid Filter Rows:	
	total_sales
▶	18591125.41

Overall KPI query results

8.2 Category-wise Sales Performance

Fruits and Vegetables generated the highest total sales among product categories.

Rank	Category	Total Sales
1	Fruits and Vegetables	2,820,059.82
2	Snack Foods	2,732,786.09
3	Household	2,055,493.71
4	Frozen Foods	1,825,734.79
5	Dairy	1,522,594.05

```
SELECT Item_Type, COUNT(*) AS total_records,
       ROUND(SUM(Item_Outlet_Sales), 2) AS total_sales,
```

```

ROUND(AVG(Item_Outlet_Sales), 2) AS avg_sales
FROM blinkit_sales
GROUP BY Item_Type
ORDER BY total_sales DESC;

```

Result Grid   Filter Rows: Export:  1				
	Item_Type	total_records	total_sales	avg_sales
▶	Fruits and Vegetables	1232	2820059.82	2289.01
	Snack Foods	1200	2732786.09	2277.32
	Household	910	2055493.71	2258.78
	Frozen Foods	856	1825734.79	2132.87
	Dairy	682	1522594.05	2232.54
	Canned	649	1444151.49	2225.19
	Baking Goods	648	1265525.34	1952.97
	Health and Hygiene	520	1045200.14	2010.00
	Meat	425	917565.61	2158.98
	Soft Drinks	445	892897.72	2006.51
	Breads	251	553237.19	2204.13
	Hard Drinks	214	457793.43	2139.22
	Starchy Foods	148	351401.25	2374.33
	Others	169	325517.61	1926.14
	Breakfast	110	232298.95	2111.81
	Seafood	64	148868.22	2326.07

Category-wise sales result

8.3 Outlet Type Performance

Supermarket Type 1 produced the highest total sales, while Supermarket Type 3 had the highest average sales per record.

Outlet Type	Records	Total Sales	Avg Sales
Supermarket Type 1	5,577	12,917,342.26	2,316.18
Supermarket Type 3	935	3,453,926.05	3,694.04
Supermarket Type 2	928	1,851,822.83	1,995.50
Grocery Store	1,083	368,034.27	339.83

Result Grid   Filter Rows: Export:  Wrap Cell Content: 				
	Outlet_Type	total_records	total_sales	avg_sales
▶	Supermarket Type1	5577	12917342.26	2316.18
	Supermarket Type3	935	3453926.05	3694.04
	Supermarket Type2	928	1851822.83	1995.50
	Grocery Store	1083	368034.27	339.83

Outlet type analysis result

8.4 Outlet Location / Tier Performance

Tier 3 generated the highest total sales, while Tier 2 had the highest average sales per record.

Location	Records	Total Sales	Avg Sales
Tier 3	3,350	7,636,752.63	2,279.63
Tier 2	2,785	6,472,313.71	2,323.99
Tier 1	2,388	4,482,059.07	1,876.91

Result Grid				
	Outlet_Location_Type	total_records	total_sales	avg_sales
►	Tier 3	3350	7636752.63	2279.63
	Tier 2	2785	6472313.71	2323.99
	Tier 1	2388	4482059.07	1876.91

Location-tier analysis result

8.5 Outlet Type + Location

Supermarket Type 1 in Tier 2 was the strongest combination by total sales, at 6,472,313.71. Supermarket Type 3 in Tier 3 had the highest average sales per record, at 3,694.04.

Result Grid					
	Outlet_Type	Outlet_Location_Type	total_records	total_sales	avg_sales
►	Supermarket Type1	Tier 2	2785	6472313.71	2323.99
	Supermarket Type1	Tier 1	1860	4302364.98	2313.10
	Supermarket Type3	Tier 3	935	3453926.05	3694.04
	Supermarket Type1	Tier 3	932	2142663.58	2299.00
	Supermarket Type2	Tier 3	928	1851822.83	1995.50
	Grocery Store	Tier 3	555	188340.17	339.35
	Grocery Store	Tier 1	528	179694.09	340.33

Outlet type + location result

8.6 Top 10 Products by Sales

FDY55 from Fruits and Vegetables was the top individual product by total sales, with 42,661.80.

Rank	Item	Category	Total Sales
1	FDY55	Fruits and Vegetables	42,661.80
2	FDA15	Dairy	41,584.54
3	FD720	Fruits and Vegetables	40,185.02
4	FDF05	Frozen Foods	36,555.75
5	FDA04	Frozen Foods	35,741.18
6	FDK03	Dairy	34,843.98
7	NCQ06	Household	34,680.19
8	NCQ53	Health and Hygiene	34,508.41
9	FDJ55	Meat	33,531.02
10	FDD44	Fruits and Vegetables	32,723.40

Item_Identifier	Item_Type	total_sales
FDY55	Fruits and Vegetables	42661.80
FDA15	Dairy	41584.54
FDZ20	Fruits and Vegetables	40185.02
FDF05	Frozen Foods	36555.75
FDA04	Frozen Foods	35741.48
FDK03	Dairy	34843.98
NCQ06	Household	34680.19
NCQ53	Health and Hygiene	34508.41
FDJ55	Meat	33531.02
FDD44	Fruits and Vegetables	32723.40

Top 10 products result

8.7 Sales by Outlet Establishment Year

The 1985 establishment-year group had the highest total sales in the analysis, at 3,633,620.15, and the highest average sales per record at 2,483.68. Results should be interpreted together with record counts.

Outlet_Establishment_Year	total_records	total_sales	avg_sales
1985	1463	3633620.15	2483.68
1987	932	2142663.58	2299.00
1997	930	2118395.17	2277.84
1998	555	188340.17	339.35
1999	930	2183969.81	2348.35
2002	929	2036725.48	2192.38
2004	930	2268122.94	2438.84
2007	926	2167465.29	2340.68
2009	928	1851822.83	1995.50

Establishment-year analysis result

8.8 Sales by Fat Content

After standardizing LF and reg, the data contained 5,517 Low Fat records and 3,006 Regular records. Low Fat had higher total sales, while Regular had a slightly higher average sales per record.

Fat Content	Records	Total Sales	Avg Sales
Low Fat	5,517	11,904,094.53	2,157.71
Regular	3,006	6,687,030.88	2,224.56

Item_Fat_Content	total_records	total_sales	avg_sales
Low Fat	5517	11904094.53	2157.71
Regular	3006	6687030.88	2224.56

Final fat-content analysis result

8.9 Top 5 Outlets by Sales

Rank	Outlet	Outlet Type	Location	Total Sales
1	OUT027	Supermarket Type 3	Tier 3	3,453,926.05
2	OUT035	Supermarket Type 1	Tier 2	2,268,122.94
3	OUT049	Supermarket Type 1	Tier 1	2,183,969.81
4	OUT017	Supermarket Type 1	Tier 2	2,167,465.29
5	OUT013	Supermarket Type 1	Tier 3	2,142,663.58

Result Grid				
Filter Rows:		Export:	Wrap Cell Content:	Fetch rows:
	Outlet_Identifier	Outlet_Type	Outlet_Location_Type	total_sales
▶	OUT027	Supermarket Type3	Tier 3	3453926.05
	OUT035	Supermarket Type1	Tier 2	2268122.94
	OUT049	Supermarket Type1	Tier 1	2183969.81
	OUT017	Supermarket Type1	Tier 2	2167465.29
	OUT013	Supermarket Type1	Tier 3	2142663.58

Top 5 outlets result

9. Key Business Insights

- The dataset contains 8,523 records with total sales of 18,591,125.41.
- Fruits and Vegetables is the highest-sales product category.
- Supermarket Type 1 generates the highest total sales among outlet types.
- Supermarket Type 3 has the highest average sales per record.
- Tier 3 has the highest total sales, while Tier 2 has the highest average sales per record.
- Supermarket Type 1 in Tier 2 is the strongest outlet-type/location combination by total sales.
- FDY55 is the top individual product by total sales in the analysis.
- Item_Fat_Content required standardization because the same categories appeared under inconsistent labels.

10. Conclusion

This project demonstrates a beginner-level SQL analytics workflow covering database setup, CSV import, data quality checks, categorical standardization, data type validation, aggregation and business-oriented exploratory analysis.

The project shows how SQL can be used to identify patterns across products, categories, outlets and locations while also demonstrating the importance of cleaning data before drawing conclusions.